01 · THE PROBLEM

LiDAR - Camera Fusion Robustness in Adverse Weather

Autonomous driving detectors fail 20 - 40% in rain, fog and snow, especially at oblique angles (90°, 180° azimuth). The degradation is angle-dependent and not well understood.

ANGLE-DEPENDENT FAILURE

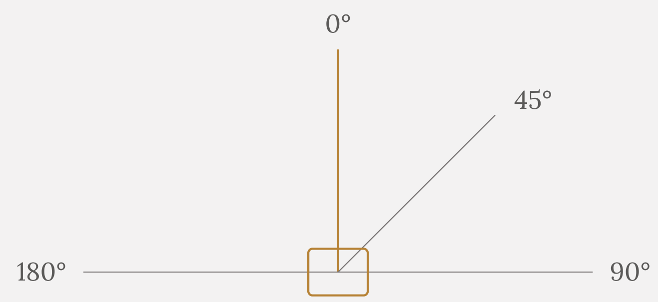
20 - 40%

detection failure in rain, fog and snow - worst at 90° and 180° azimuth.

RESEARCH QUESTION

“Can Combined fusion outperform single models at extreme angle–weather combinations?”

AZIMUTH



- 0°, 45°, 90°, 180° azimuth
- Clear, rain, fog, night
- Oblique angles (90°, 180°) degrade worst

02 · SOLUTION

Phase 1: measure how accuracy varies, using pre-trained models

Using pre-trained 3D object detection models to measure how detection accuracy varies by angle, weather and dataset - then combining two pre-trained models.

PHASE 1 · WHAT VARIES

Angle, weather, datasets

- **Angle** - 0°, 45°, 90°, 180° (azimuth)
- **Weather** - clear, rain, fog, night
- **Datasets** - nuScenes (primary), Ithaca365 (validation), CADC (snow)

MODEL 1 · ECCV 2024

SAMFusion - good for bad conditions

Deep learning model designed specifically for adverse weather.

- Explicitly trained on rain / fog / snow scenarios
- Sensor-adaptive: learns which sensor to trust when
- Pre-trained weights available (Princeton CI Lab)
- Robust in bad weather

MODEL 2 · ICRA 2023

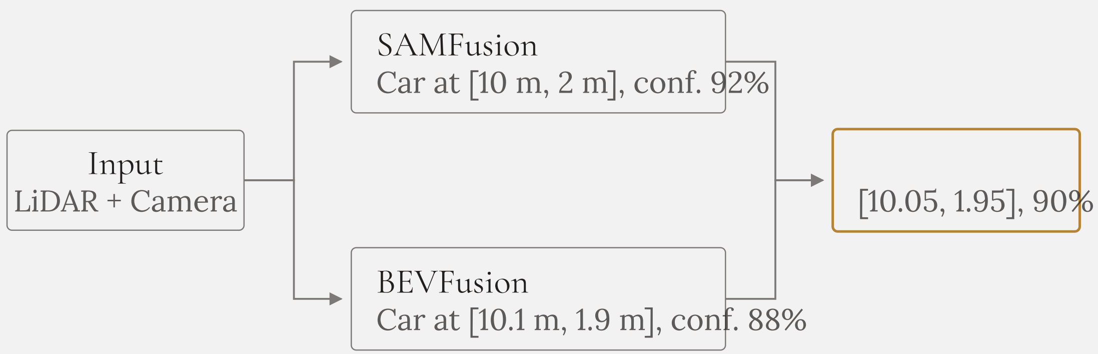
BEVFusion - accuracy

Industry-standard 3D object detection model.

- Production-proven
- High accuracy in clear weather
- Pre-trained weights available (NVIDIA NGC)
- Pre-trained on nuScenes clear weather

03 · STRATEGY

Run both models, combine their predictions



SCENARIO	SAMFUSION ALONE	BEVFUSION ALONE	COMBINED Combined
Clear weather	Good	Excellent	Excellent
Heavy fog + 90° angle	Excellent	Poor	Excellent
Rain + 45° angle	Excellent	Good	Excellent

METHOD

Weighted by weather

- Bad weather → 70% SAM, 30% BEV
- Good weather → 50% SAM, 50% BEV

METHOD

Confidence

- Weight by each model's own confidence score

COMBINED PROCESSING

- Run SAMFusion on each frame
- Run BEVFusion on each frame
- Combine predictions
- Bin by angle + weather
- Calculate combined mAP per bin

04 · DELIVERABLES

Heatmaps: angle × weather → mAP

Question: does the combined model outperform both in adverse weather and at high angles?

Heatmap 1 · SAMFusion alone | Heatmap 2 · BEVFusion alone | Heatmap 3 · Combined

	0°	45°	90°	180°
CLEAR	mAP	mAP	mAP	mAP
RAIN	mAP	mAP	mAP	mAP
FOG	mAP	mAP	mAP	mAP
NIGHT	mAP	mAP	mAP	mAP

Cells are filled per bin by the Phase 1 runs — select a heatmap above.

PHASE 1

- Systematic characterisation using the combined approach
- Three heatmaps: SAMFusion alone, BEVFusion alone, combined
- Cross-dataset validation
- Statistical analysis: does the combined model outperform both in adverse weather and clear conditions?
- Answer: here is how the combined approach handles different angles

PHASE 2

- Optimisation of combined weighting (if time allows)
- Real-world deployment considerations

Five options for CANBUS integration

OPTION	WHAT IT DOES
1 · Input only	Read vehicle speed, steering angle, acceleration to provide context for detection - improves object classification by understanding vehicle motion.
2 · Output only	Send brake commands to trigger AEB based on detection confidence: detection confidence > threshold → activate AEB via CANBUS.
3 · Full AEB loop	Read vehicle state, then decide AEB trigger based on detection + speed + angle + weather: read speed, brake status and steering from CANBUS; detect objects, confidence, angle and weather; decide the AEB threshold from all factors; write the brake command via CANBUS.
4 · Validation	Use CANBUS data to validate detection logic: braking → detected objects stable ahead; accelerating → detected objects move backward.
5 · Contextual weighting	Use vehicle speed to adjust SAMFusion vs BEVFusion weights: low speed / bad weather → favour SAMFusion; high speed / clear weather → favour BEVFusion.

06 · CONCLUSION

Combining a robustness specialist with an accuracy specialist

- **SAMFusion** - weather robustness specialist
- **BEVFusion** - accuracy specialist

RESEARCH QUESTION

"Can Combined fusion outperform single models at extreme angle-weather combinations?"

PHASE 1 ANSWERS

- Three heatmaps plus cross-dataset validation
- Statistical analysis of the combined approach against both models
- How the combined approach handles different angles

COMPLETE TECH STACK

Languages & frameworks

- Python 3.10+
- PyTorch 2.0+
- MMDetection3D
- ONNX

Processing & analysis

- Pandas
- NumPy / SciPy
- scikit-learn

Dataset tools

- nuScenes DevKit
- KITTI Tools
- Open3D
- CARMAKER

Visualisation & tooling

- Matplotlib
- Plotly
- Jupyter
- VS Code, Git